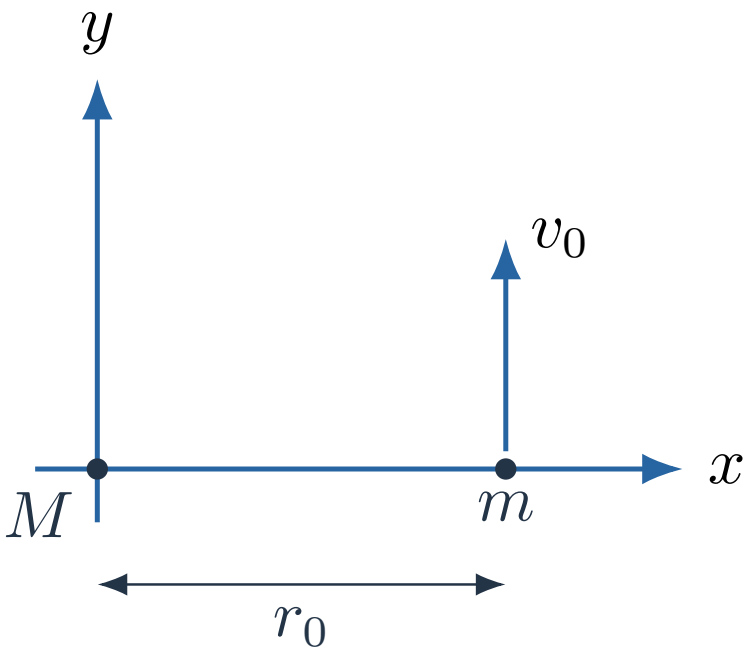
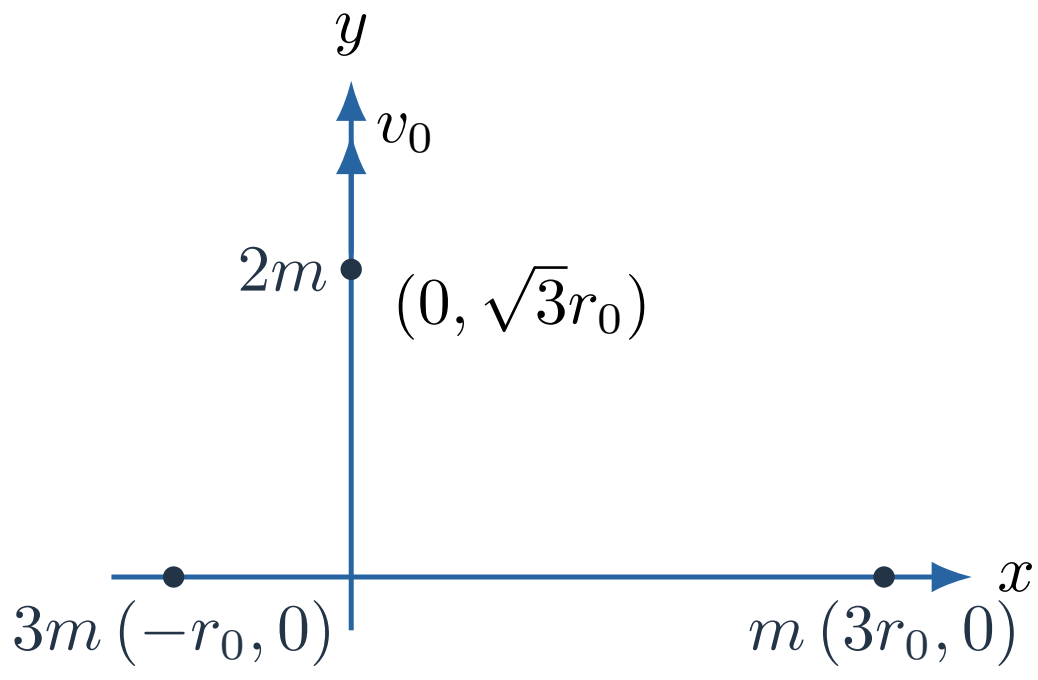
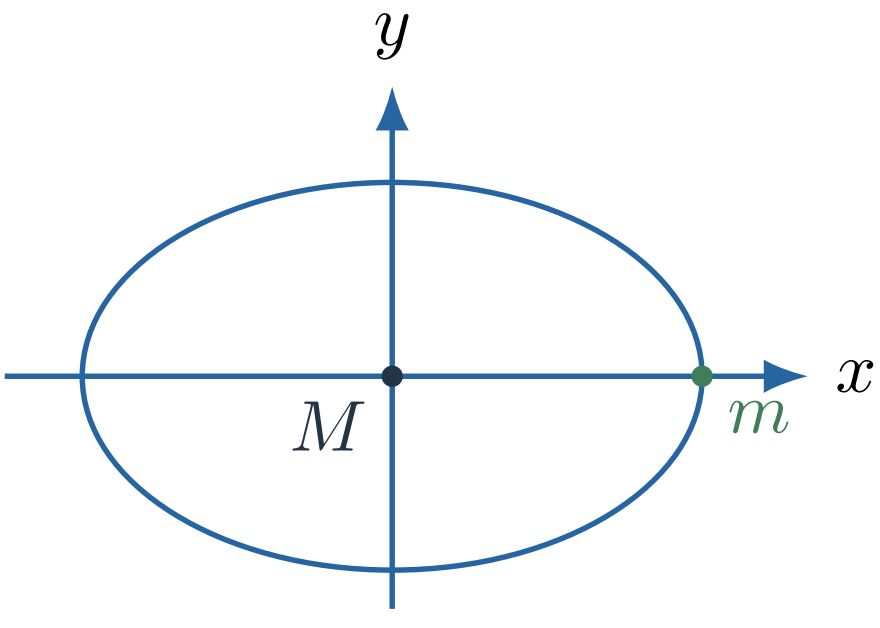


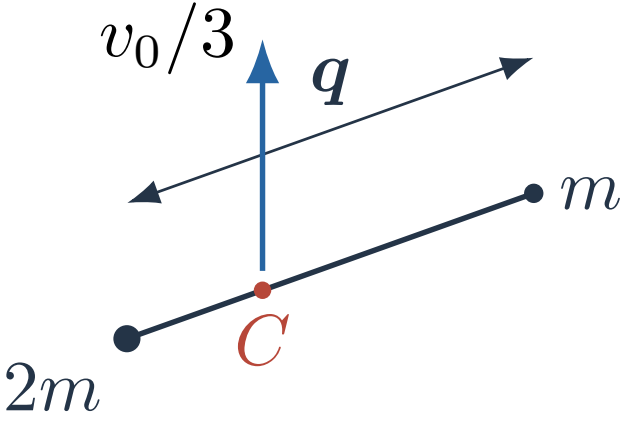
(a)



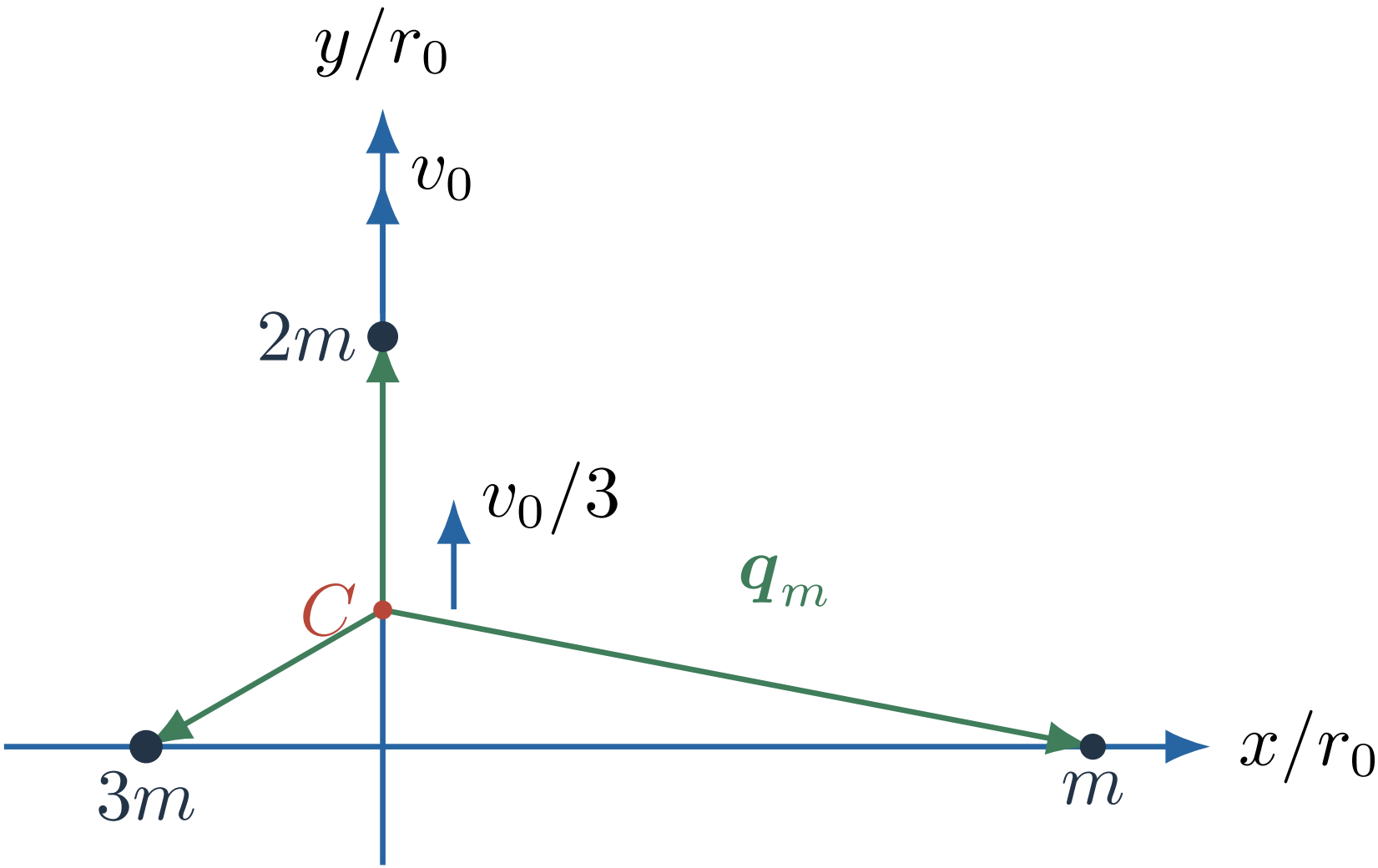
(b)



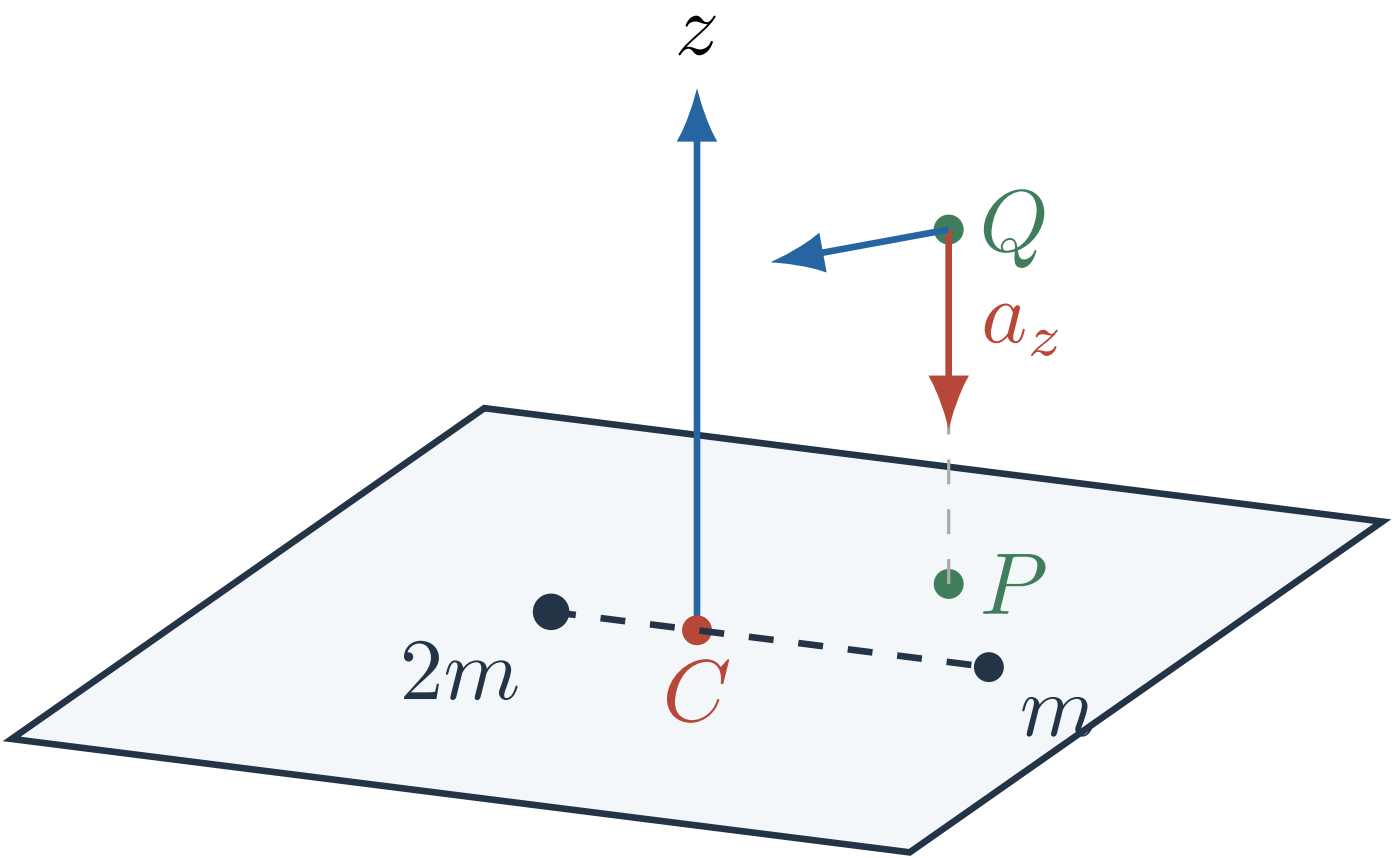
r_0 与 v_0/ω_1 为半轴



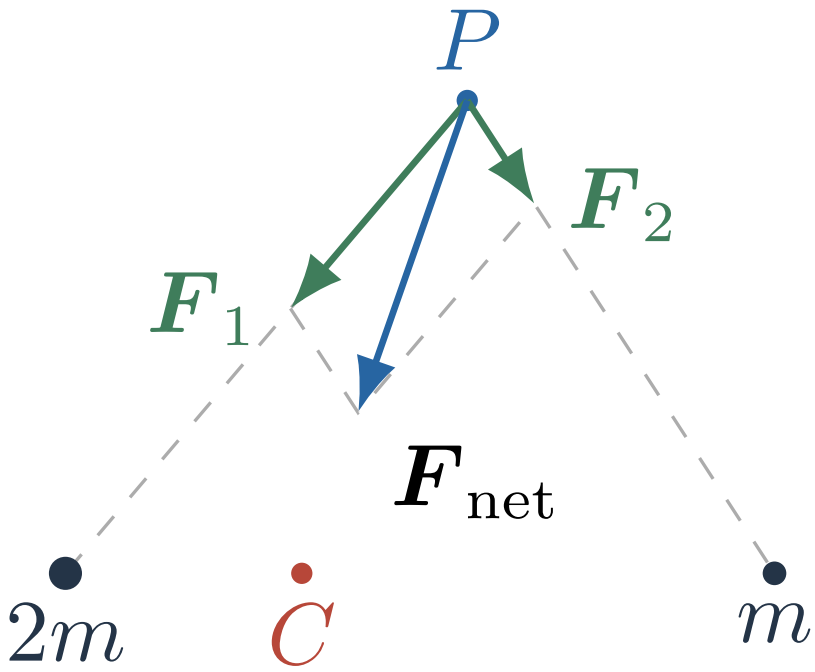
$$\boldsymbol{r}_m = \boldsymbol{C} + \tfrac{2}{3}\boldsymbol{q}$$



$$\omega_3 = \sqrt{6km}$$



平衡集合：$z = 0$；质量以 $2m, m$ 示意



$$\boldsymbol{F}_{\text{net}} = -k\, m_P (3m)(\boldsymbol{r}_P - \boldsymbol{C})$$